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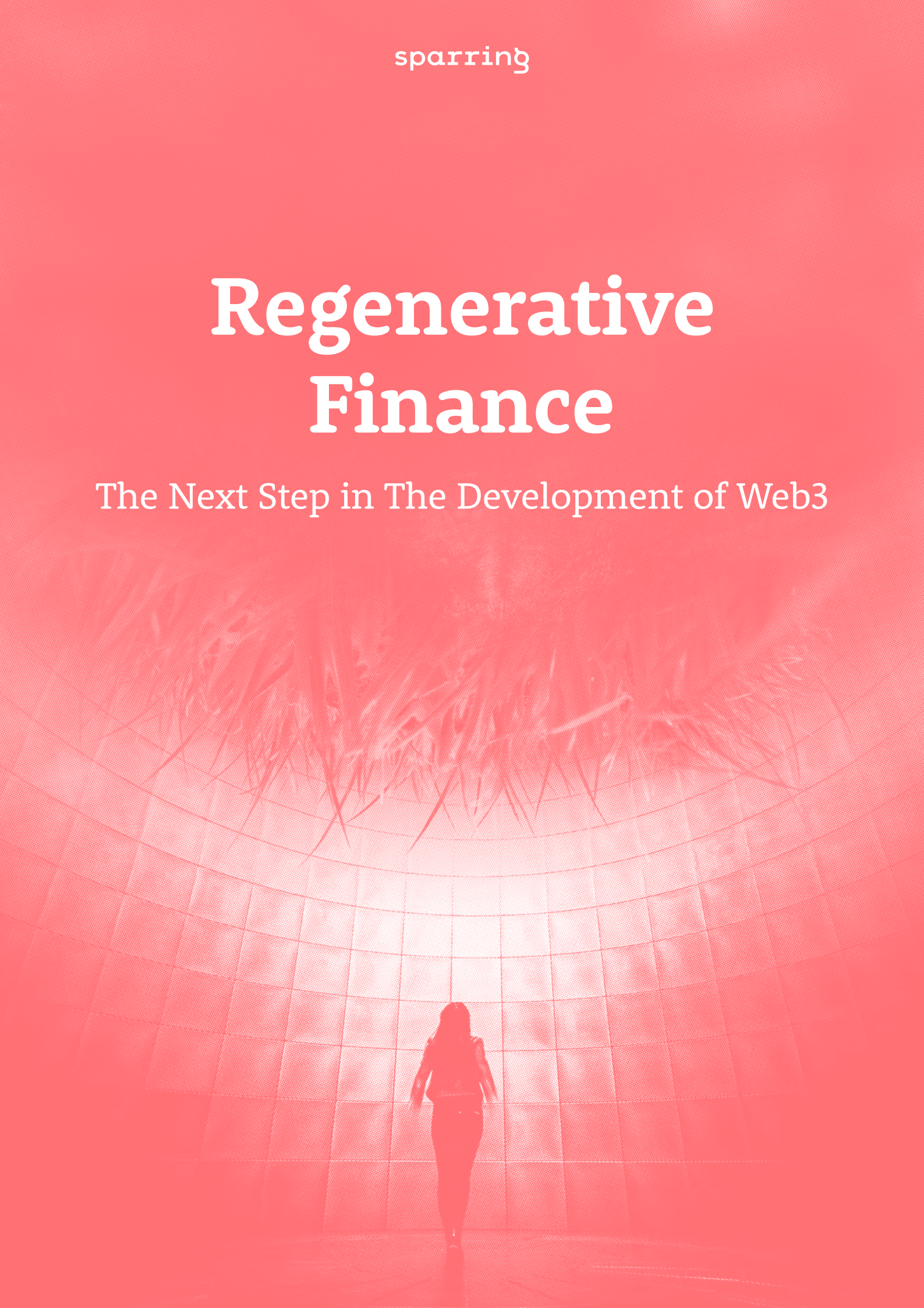
# Regenerative Finance

## The Next Step in The Development of Web3



**Chapter jump:**

Impact Investing | DAOs as the Future of Regenerative Organizations | Circular Economy and Degrowth | Community Currencies, Time Banks, and Sustainability | Co-ops and Supply Chains | Decentralized Energy

**The concept of regenerative finance or "ReFi" refers to the process of using financial markets to correct the problems that the markets themselves have created. Economic systems built on the Web3, cryptocurrencies, and digital tokens can help start the regeneration of the world economy. In many ways, these are the freest market systems to date thanks to ReFi's transparency and democratization of the use of resources.**

# Closer to Nature

The term "degen" (an abbreviation of "degenerate" or unnatural) has become a common term in the cryptocurrency community. Its use stems from the high-risk, high-reward nature of digital asset trading, as well as people's propensity to bet on big profits. As a reaction to this mentality, where the only value is monetary income, comes the term "regen" (an abbreviation for "regenerative") which refers to a return to the natural essence of things.

The phrase "degen to regen," or from unnatural to natural, became a mantra of this movement. The goal of this trend is to create a financial system that promotes positive social and environmental values through healing and reusing natural resources and ecosystems. Without generating unnecessary waste, shifting away from excessive consumption, and ultimately focusing on the general well-being of individuals, society, and the planet as a whole. It is a way of thinking about the economy that considers the negative externalities that traditional capitalism has often ignored. These include, for example, environmental destruction, social inequality, and financial instability.

The transformational power of Web3, which decentralized nature has the potential to unlock new economic values and build a fairer future, should also help ReFi's goals. In its simplest form (i.e. in the form of decentralized finance - DeFi) it promotes the democratization of access to financial services and the elimination of middlemen, who often abuse their position to accumulate resources. However, even DeFi still has profit as its main value, and therefore it is necessary to

look further to systems that place emphasis and value on the creation of positive externalities, such as ReFi.

ReFi is an emerging field that can encompass a wide range of activities such as the use of impact investing, community development financing, green bonds, crowdfunding and other innovative financial instruments.

ReFi aims to align the financial system with regenerative principles to support the transition to a more sustainable and resilient economy, while simultaneously addressing social and environmental issues. ReFi can also include areas such as regenerative agriculture, conservation finance and the use of blockchain technology to support sustainable and transparent supply chains. The concept of a regenerative economy is still in its infancy, but several examples of Web3 projects that use ReFi principles to create positive change already exist.

## Carbon Trading Using Tokens and NFTs

In the last few decades, we have seen with our own eyes the devastating effects of carbon emissions and the pollution of our planet with plastic. At this point there is no longer any doubt that current human activity and climate change are related. What people have caused; people should fix. To offset some of the damage caused by industrialization, various carbon trading systems have been created. These systems allow companies to offset their emissions by investing in projects that remove carbon from the atmosphere. However, the current major problem with most carbon trading systems is that they are often non-transparent and untrustworthy. This is where Web3 comes in as one of its primary uses is ReFi.

By using so-called non-fungible tokens (NFTs), it is possible to create a decentralized carbon trading system that is more transparent and less prone to abuse. Companies can take advantage of Web3 and blockchain technology to tokenize carbon credits and make them transparently available on a publicly distributed ledger to support responsible carbon offsets. An example of such a project is UPCO2, which is the first tradable token containing a digital version of a carbon credit. The above solutions not only facilitate the tracking and verification of carbon credit transactions, but also open the market to a wider range of participants, a market which has been very limited until now. At the same time as more companies will be forced to offset their

emissions, which will likely increase demand for these tokens, thus leading to greater investment in carbon-reducing projects.

## Other Areas of ReFi

With the growing awareness of ReFi and the increasing number of projects that start to apply its principles, the spectrum of its use is expanding.

One area that is booming is impact investing or investing with a positive or regenerative impact. This is a type of investment that not only has the goal of achieving financial returns but stresses the importance of positive social or environmental impact above all else. Thanks to ReFi, it is also possible to create regenerative DAOs, i.e. decentralized autonomous organizations, with the goal of investing in projects with a positive impact on our planet. Web3 and decentralized autonomous organizations provide a new level of transparency and accountability that ensures fair distribution of funds among all stakeholders. An example of such an organization is the philanthropic non-profit Big Green DAO.

Another area where ReFi will have a significant impact is community governance. In many contemporary communities, decisions are made by a small group of people, often in a centralized manner. Thanks to ReFi, it is possible for everyone to participate in the management of communities where they invest their time, property, or finances. This will lead to more democratic and effective decision-making as well as improved social relations. Traditional Dream Factory is an example of one of these communities, as it is one of the first prototypes of a new form of living in harmony with nature through a regenerative eco-village. The eco-village is managed as a DAO using a digital token that includes, in conjunction with community membership, the right to use the village's premises and facilities, along with the right to co-decide its direction and development.

**As Web3 becomes more and more mainstream, we have a unique opportunity to use its new technologies to build a better, fairer, and more sustainable society that not only pursues its own goals, but also overall benefits for nature and the planet, as well as the ecosystems that exist within them. So, let's move on from degens to regens!**

# Impact Investing and How it Differs from ESG

## What is Impact Investing

Impact investing is a strategy that seeks to create a specific positive externality or outcome - that is, a measurable positive social or environmental benefit. It differs from philanthropy, such as donations, because this investing involves an expectation of financial returns that are at least comparable to market returns.

The categories of intended impact mainly include those that are (i) social, such as access to education or employment opportunities, (ii) environmental, such as reforestation projects or so-called clean energy initiatives, or (iii) individual, such as increased personal access to medical care, clean drinking water, or healthy agricultural and food solutions.

Impact investing is not limited to any particular asset class such as stocks or bonds. It can include private equity investments, project bank financing or microfinance loans, to support small business ventures, for example in countries with insufficient bank financing. It is also often carried out by retail investors through funds, but direct investments in projects or microfinance initiatives are also common. This can also include crowdfunding or investing through new investment vehicles such as decentralized autonomous organizations (DAOs).

A good example is Triodos Investment Management, a subsidiary of Triodos Bank based in the Netherlands, which manages more than a dozen sustainable investment funds. Triodos has been actively involved in positive impact investing since 1995 and holds assets of approximately 5 billion dollars. Their main areas of investment interest include renewable energy, food and organic agriculture, healthcare, and education. It is also one of the founding members of the Global Impact Investing Network.

This type of investing can mean many different things and can be characterized in different ways. However, to qualify as positive impact investing, the following two basic conditions must be met:

1. Positive impact is clearly formulated, described and measured; and
2. an expected financial return comparable to conventional types of investment must be intended.

An example of a positive impact investment could be the following:

Let's say you live in a densely populated and expensive urban center in dire need of affordable housing. Suppose that (a) you have surplus capital and (b) you want to use some of this capital to support affordable housing.

There is a local developer who offers you the opportunity to invest in a project which proceeds will support the construction of affordable housing. This project is expected to generate market (or better) returns. This is impact investing because there is both an intended positive outcome and a clear expectation of financial return.

## Is Impact Investing the same as ESG?

Impact investing creating positive externalities is not the same as ESG (Environmental, Social and Governance) requirements, i.e. requirements for sustainability of companies, businesses and organisations in the areas of social and environmental responsibility. Although there are certainly some overlaps.

ESG is a certain analytical framework that helps stakeholders understand and manage risks and manage risks and opportunities within an organization (or even an investment portfolio). And while this framework also applies to investment decisions or analysis, it is not the same as using funds with the explicit intention of creating a positive externality or impact.

Investment decisions that consider ESG sustainability requirements may not yet aim to generate any specific positive externalities. ESG-conscious investors operate under the belief that business companies that perform better on environmental, social or governance issues are simply better

investment opportunities. This has already been proven by several market analyses. These companies are more aware of specific risks and can mitigate them accordingly. They are also better prepared to take advantage of opportunities arising from changes in social norms or the reality of our environment.

Many investors focused purely on ESG metrics also care about the values of environmental protection, social behaviour and making sure their investments have a positive impact on the world, but their capital allocation decisions are much more influenced by managing risks and opportunities than explicitly creating positive externalities.

# Conclusion

Whether you are, or want to be, a positive impact investor or an ESG enthusiast, you probably believe in two important principles. These are:

1. Completely uncontrolled capitalism in its current form does not work for all stakeholders. This is because the constant pressure for growth and competition creates a toxic environment in terms of social relations and environmental impacts.
2. Companies seeking capital owe the market more transparency and accountability for what they are doing to promote positive change.

Back in 2018, Larry Fink, CEO of Black Rock, the world's largest investment asset management firm, wrote the following in a public report: "For any business to thrive over the long term, it must not only deliver financial results, but also demonstrate how it is making a positive contribution to society as a whole."

**Real wealth and value are not just money in the bank. Wealth must be defined and managed with keeping the well-being of all individuals, their environment, and the society in mind. This can only be achieved by harmonizing multiple types of wealth or capital, beyond ordinary financial, material, and technological capital. This includes, above all, social and relational wealth or capital, cultural capital and capital for individual well-being and satisfaction.**

# DAOs as the Future of Regenerative Organizations

As decentralized autonomous organizations (DAOs) proliferate across the new Web3 ecosystem, new types of DAOs are emerging. The new domain in which they operate is referred to as "Impact DAOs", or Decentralized Autonomous Organizations with positive impact, which create positive externalities for the ecosystems in which they operate.

These decentralized organizations extend the original DAO model using Web3 technology to have a net positive, sustainable and regenerative impact. At the same time, these projects are often experimenting, innovating rapidly, and redefining how public goods are created and how resources are used and shared.

DAOs are using smart contracts, tokenization and basically new technologies and new governance models to incentivise actions that regenerate us and our planet. These organizations offer us opportunities to address complex societal problems and create sustainable ways of working together that add value in themselves and motivate regenerative behaviour.

## Principles of Regenerative DAOs

Organizational structures are built from the ground up according to the wishes of the community. In practice, DAOs often operate as cooperatives and use the principles of holacracy or sociocracy to manage themselves. Once a common goal is decided, members of these organizations often form so-called circles or other ad-hoc working groups that coordinate around their defined priorities. Coordination can be challenging because there is no top-down management, which means that good flat organizational structures are particularly important for DAOs. Members need to be motivated by them to work together.

In addition to these basic operating principles, DAOs with positive impact have several unique characteristics:

1. **They leverage network effects to combine their efforts and achieve greater impact**

   DAOs leverage the efficient allocation of talent, energy, and resources where they are needed most. This allows these organizations to connect with each other and leverage larger structures and a wider range of contributors when needed to solve societal problems.

   The community first decides collectively what values, goals or activities it will participate in. Then, for example, members who focus on agriculture can use Regen Network organizations to learn about best practices for regenerating specific areas and to better understand how to get the most ecological and economic value from regenerative transformations. Alternatively, geo-sensors (such as those being built by Astral Protocol) and smart contracts can be used to automatically reward positive ecological behaviour. Or, as the DAO community begins to gain momentum and real-world impact, it can use the quadratic funding platform Gitcoin to further support projects that are central to such a community.

2. **Regenerative design**

   Impact decentralized autonomous organizations are specifically designed to have a regenerative and sustainable impact on the ecosystems they co-create. Because resources increase over time in regenerative systems, using community digital tokens can be used to systematically make wealth and resources circulate back into the community. And its members can then collectively vote and suggest ways to reuse them in a way that maximizes their subsequent positive impact.

3. **Positive externalities**

   DAOs with a positive impact often have a sustainable and long-term strategy to achieve positive results in their area of operation. Web3 technology is a great help to this. Tokenization allows our resources, capital and assets to take on a virtual life using

blockchain technology and be leveraged within organizations so that their true impact can be measured. Projects focused on carbon offsets, cultural capital and scientific research are just a few examples of these new organizations.

## Case Studies

Regenerative DAOs are currently mainly used for social and positive climate change projects. These organizations can act as building blocks on which other similar organizations can build. The following are some examples of such organizations:

The Regen Network is a decentralized positive impact organization that is exploring how to link the value of ecological health with economic value, and is creating an open community-driven registry of carbon and environmental assets. It is a community of scientists, researchers and project developers creating a system of quality tokenized carbon credits. This community has collectively created more than 40 methodologies for 12 million hectares of ecological restoration sites.

Molecule offers a digital marketplace with scientific research and scalable frameworks and processes for building biotech DAOs. Molecule's long-term goal is to decentralize the processes of ownership, financing and management of drug development. Molecule enables communities of patients, researchers and investors to directly fund, manage and own research-related intellectual property. Tokenizing intellectual rights at their early stage, along with research projects, enables to gain necessary resources to create breakthrough biotechnologies.

Metropolis (formerly Orca Protocol) creates DAO management tools that put their members at the center. This project allows DAO members to coordinate within so-called pods (organizational mini-structures). Pods are small working groups organized according to the expertise and ability of their members to contribute to the overall organization. They allow these mini-groupings within larger structures to integrate and help achieve common goals. The goal of the Metropolis project is to create a variable-complexity governance framework that can act as a kind of connective tissue for the management of decentralized organizations.

## Challenges

Although the positive future of decentralized organizations is clear, some related major issues need to be addressed. Steering its members towards real regenerative outcomes, working together without traditional hierarchies, and authentically verifying the impact of their activities are some of the obstacles that DAOs face. As the space of decentralized organizations itself is still emerging, new solutions and trade-offs will need to be sought in the future to effectively meet their goals while working towards effective use of Web3 technology.

# Circular Economy and Degrowth:

### Change Towards a Sustainable Future

The concepts of circular economy and degrowth primarily challenge the traditional capitalist linear economy that underpins our society today. Both new approaches explore alternative ways of allocating resources to realize a sustainable lifestyle for humans and their communities that is in harmony with the principles and laws of nature.

Although both approaches have their own distinct strengths in different aspects of our modern socio-economic society, they also share common elements and features that predestine them as important tools for overcoming the current unsustainable economic, social and ecological crisis.

**Circular Economy:**

The circular economy focuses on continuously optimizing the use of resources through the redesign of distribution systems, processes, as well as the products themselves, aiming to minimize waste.

It is an economic model that seeks to minimize waste and resource intensity by trying to keep products, materials, and resources in circulation for as long and as efficiently as possible. The aim is to replace the linear economic model that focuses only on production - use - disposal mechanic.

The circular economy seeks to achieve the above goals by thinking about the entire life cycle of a product - from its design to its final processing. This model promotes the use of renewable resources, waste minimisation and recycling of materials. For example, it uses sharing, repairing, recycling, and remanufacturing, all with the aim of reducing the negative environmental impact of production and consumption.

The circular economy also offers opportunities for businesses to reduce the cost of resources while increasing the efficiency of their use. The model also encourages the creation of new markets for recycling and recovery, thereby promoting new business opportunities.

**Degrowth:**
Degrowth is a concept that seeks to challenge the traditional economic model that focuses on constant growth and increasing GDP as a measure of success. Degrowth means looking for an alternative economic model that aims to reduce resource and energy consumption, focusing on sustainability, quality of life and sharing instead of accumulating wealth.

This concept assumes that the current economic growth is unsustainable in terms of resources and energy and that the current model of our society leads to increased social and environmental inequalities. However, it does not mean rejecting economic growth as such, but rather replacing it with sustainable ways of living and managing natural resources. This concept focuses on the promotion of local economies, renewable energy, sustainable production and consumption, the use of sharing and cooperation and the promotion of social welfare.

## The Path to Systematic Change

However, simply incorporating circularity or a degrowth economy may not be enough to ensure a truly sustainable future. Realizing meaningful change requires a coherent and systemic transformation that incorporates the key principles of both approaches. Together, circularity and

degrowth provide a blueprint for shifting social, economic and environmental paradigms, using the following common principles:

- Prioritizing long-term social well-being over profit maximization,
- promoting collaborative consumption and the sharing economy,
- promoting of diverse, locally focused economic models, and
- supporting adaptation and transformation to resilient and regenerative communities.

**For the field of fintech, we can show two examples of such models:**

- Ethical banks - https://www.triodos.co.uk/, https://www.nerdwallet.com/article/banking/socially-responsible-banks
- Financial cooperatives, producer and consumer cooperatives and decentralized supply chains - https://www.skupina.coop/en/170-let-spotrebnich-druzstev-skupiny-coop

**Other areas may include:**

- Urban Food Gardens - https://www.metrofarm.cz/en/
- Alternative Food Networks which include community supported agriculture (CSAs), farmers' markets, fair trade, urban agriculture, specialized forms of organic agriculture, direct farm retail, and the slow food movement.
- Eco-communities and eco-villages - https://spoluzemevrabsko.cz/home/, https://www.zemesouzneni.cz/en/, https://www.traditionaldreamfactory.com/
- Community Currencies - https://en.wikipedia.org/wiki/Chiemgauer
- Barter, Swap and Circular Markets - https://www.awesomefoundation.org/en/projects/66523-barter-market
- Hacklabs - https://www.paralelnipolis.cz/en/ , https://brmlab.cz/
- Decentralized forms of renewable energy generation and distribution - https://www.unescap.org/sites/default/files/14.%20FS-Decentralized-energy-system.pdf

# The Change Is in our Hands

People must abandon the illusion that infinite growth is possible on a finite planet. Circularity and degrowth are both a vision and a path to environmental security and social abundance. By embracing principles such as reducing waste, increasing resource efficiency, and striving for fair

values - individuals, businesses and governments can move towards a sustainable and inclusive future.

The intersection of circularity and degrowth presents a unique opportunity to redefine our values and goals that will lead us to a sustainable and equitable future. As we stand at a critical juncture, it is essential to recognize the need for holistic change that embraces both approaches in our daily lives, social and decision-making processes.

By embracing the complementarity of these concepts and actively integrating them into our personal decisions, policies, and industrial practices, we can lay the foundations for a resilient future that prioritizes our collective well-being, environment, and prosperity.

# Community Currencies, Time Banks, and Sustainability

**Community currencies and time banks are new and innovative approaches to sharing resources and strengthening communities. These alternative economic systems focus on using our skills, time, and resources to benefit all members of society. They offer a wide range of benefits for individuals and the community by increasing social interaction and trust between people, strengthening local economies, and contributing to sustainability.**

Community currencies are a tool for strengthening local economies and creating social cohesion. These currencies are often used in specific geographical areas and allow residents to buy and sell goods and services only among themselves. This promotes local entrepreneurship and creates space for sustainable economic development.

Time banks work on a simple principle - instead of a traditional currency, they use time as a measure of value. Each member can offer his or her skills and time to other members of the community and in exchange receives time credits. These credits can then be used to get help, services, or products from other members.

After the global financial crisis in 2008, it became clear that conventional public money is undemocratically managed and not designed to work in the interests of all. Within the current globalized capitalist economy, there is a growing critique that current financial and monetary institutions (i) are at the root of recurrent economic crises, (ii) encourage unethical and risky economic behaviour such as speculation in markets, (iii) marginalize entire communities by restricting their access to financial resources, and (iv) promote an ecologically destructive economic growth imperative. All this is done solely in the interest of capital accumulation.

## Community Currencies

One response to the above developments is community currency. What is it? It is a special local currency that can only be used in a specific locality in shops or other businesses that have committed to accepting it as an alternative to the official national currency and encourage spending in the local economy. This currency can cover either an entire city, town, or neighbourhood or, as in the case of a particular community currency, the Brixton pound, just one London borough. The Brixton pound, the UK's largest community currency, is used by around 500 businesses and can be purchased at an exchange rate equivalent to the national British pound.

Although each community currency reflects the realities of its locality, the common element is always to support the local economy in such a way that local currency, which is only of value within a given community, circulates only within that community and does not flow out beyond its borders. The main concern is therefore to maintain local money and therefore local resources and value. Numerous studies have shown that local independent retailers return much more money back to their communities than large retail chains.

While some community currencies must be bought at local issuing points with national currency, others give citizens the opportunity to 'earn' their income through activities such as charity work - which helps to explain why these currencies often flourish in areas and at times when many

people have been out of work.

> "There is a strong correlation between unemployment and the size of complementary currencies," says Christian Gelleri, a German economist who founded a community currency called the Chiemgauer, which can be used in Bavaria, Germany. "One successful example in 2019 was Sardex in Sardinia, which has more than 3,000 businesses and a turnover of almost €50 million ($54.7 million, £44.4 million). The unemployment rate there was 15%. In the Chiemgauer currency, we had a turnover of €6.3 million, with an unemployment rate of 1.9%."

Local or community currencies are not rare in financial history and have taken many different forms. Some of these include the local exchange trading system (LETS), which consists of a system of mutual credit charged within a specific community, and time banks, in which people use time-measuring vouchers to exchange services. These are institutions that are not regulated by market or state institutions, but by the social relations of the community. Although they are issued by non-state civic entities such as local NGOs and civic associations, they sometimes also work with local governments.

This phenomenon is also gaining importance in Europe, for example in Amsterdam, where a neighbourhood local currency was initiated by the residents of the Oost district, where the Makkie currency was established in 2012. In the Noord district, the time bank Buuro has started its operations and in the Zuid-Oost district they are developing their own digital currency using blockchain technology. These projects have also recently received official support from the city. In the context of its strategic plan made for each of the city's districts, there will be a greater emphasis on building community wealth and working on the development of a new digital currency.

In the Czech environment, the Corrency project, a community currency system to support the local economy, is a great example of this, allowing Czech town halls to set clear and simple rules for the efficient and transparent use of funds to support residents or selected business areas. Local communities realize that if resources and wealth circulate and are not lost to neighbouring markets, the whole region will benefit from such an initiative.

# Time Banks

Many communities that are structurally excluded from the global economy, such as communities in remote and rural areas, refugee camps, urban villages, and slums, exchange their most valuable asset - time - with each other, instead of money. Time, which is allocated relatively equally to each human being, can be tokenized as a community currency that facilitates local transactions. Whether it is gardening, medical care, transportation, minor household repairs, computer help, picking up groceries, or cooking a meal, these exchanges of human capacity are of equal value. Their value is based only on the time that is devoted to them.

For example, major Hong Kong NGOs are working on a joint citywide time bank project to digitize and tokenize time vouchers and scale a time exchange platform, supported by cloud and blockchain technologies. During the COVID pandemic, many people joined the program to exchange health-related goods and services.

Another example is the Kola Nut Collaborative time bank in Chicago, which has facilitated more than 1,000 transactions to date. There are approximately 500 time banks in the US, involving nearly 40,000 members of their communities in local exchanges. "Participating in a time bank or other form of mutual aid is one way we can meet our needs without exchanging money, reducing the need to always turn to outside markets first," explained its founder.

# Impact on Sustainability

Community currencies and time banks contribute to sustainability by promoting local economic resilience and social cohesion. Community currencies encourage local economic activity and support local businesses, reducing the carbon footprint associated with long-distance trade. Time banks facilitate the exchange of services, support the sharing economy, and reduce waste by utilizing existing resources and skills within the community. These initiatives foster a sense of belonging, promote sustainable practices, and enable individuals to take an active role in building more resilient and environmentally conscious communities.

# Financial Co-ops, Producer-consumer Co-ops and Decentralized Supply Chains as Instruments of Sustainability

**Financial co-ops, producer and consumer co-ops and decentralized supply chains are important elements of the economic system that bring significant benefits and support the sustainable development of society.**

## Financial Co-ops

Financial co-ops are communities of people who have come together to offer financial services to their members. They are based on the principles of mutuality, solidarity, and democracy. They give individuals and small businesses access to credit, savings and other financial services that might otherwise be unavailable. In this way, financial co-ops promote inclusion and help improve the economic situation of communities.

Financial co-ops have open membership and, unlike banks, may be more concerned with the financial health of their members than with making a profit. The control of the co-op is democratic, with each member normally having one vote. Their individual financial situation or level of ownership of the co-op is typically irrelevant.

The members of the co-op are both owners and customers. The size of the co-op is based on the number of participating members. With more members, a financial co-op has more resources to offer financial products, lower fees, lower interest rates on loans and higher returns on savings. In addition to the financial products and services a co-op offers, they can also be a source of financial education for their members and others. Services that co-ops provide may include retirement planning or financial literacy.

One of the most common examples is agricultural co-ops, which bring together individuals and

companies involved in farming and food production. A credit union of this type will include basic banking services, mortgages, business loans, educational and other training opportunities, and general networking opportunities.

## Producer and Consumer Co-ops

Producer and consumer co-ops are groups of people who have come together to jointly produce, buy, and sell goods and services. These co-ops create an environment in which it is more profitable and fairer to trade. Producers can share the cost of production and achieve greater efficiency, while consumers gain direct access to high quality products and services at fair prices. Producer-consumer co-ops thus support a sustainable business model and create space for cooperation and sustainable community development.

For example, COOP is an internationally used brand that brings together 46 consumer co-ops operating around 2 500 stores. COOP started in the mid-19th century in the Manchester area of England and quickly spread to many countries around the world, including the Czech Republic. In Switzerland, Italy, Finland and Slovakia, it is the market leader in food retailing and also has a strong position in Scandinavia, the UK, Japan and other Asian countries.

Another example is the Park Slope Food Coop, a member-owned and operated grocery store founded in 1973 in the Brooklyn borough of New York City in the United States. This store defines itself as an alternative to profit-driven businesses and, as a co-op, integrates a community of organic food supporters. It promotes non-toxic organic farming and tries to avoid selling processed products. This store sells food only to its members and they contribute their time, up to about 3 hours per month, to the daily tasks of maintaining and operating the store.

## Decentralized Supply Chains

Decentralized supply chains are an alternative to traditional centralized models of management and distribution of goods. These chains allow individual links in the supply process to have greater autonomy and responsibility. Product and transaction information is transparent and stored in a decentralized manner, which increases trust and reduces the risk of manipulation. Decentralized supply chains thus promote a more efficient, secure, and sustainable flow of goods

from producers to consumers.

Decentralization can be achieved using blockchain technology, which is an immutable digital ledger of transactions and recorded information. It is possible to envision a large spreadsheet to keep track of all goods and a registry and inventory system to record, track, monitor and transact them.

The blockchain reflects a faithful representation of reality at a given point in time, thereby creating trust between parties. The use of blockchain technology allows to overcome the current limitations of food supply chain management by providing better transparency and security. Information about product movements and sources are recorded on a public and immutable registry, which helps eliminate fraud and mistrust. It is also possible to trace the origin and journey of products from their origin to the final consumer. This enables better quality control, compliance with standards and consumer protection.

Rujamar, Spain's leading producer of cage-free eggs, has incorporated blockchain technology into its production process with an initial investment of € 235,000. The first phase of the transition involves medium and large eggs on a production line with 120,000 hens. And their outlook for the future is truly inspiring: Rujamar plans to expand the system to its entire production, i.e. 1.2 million animals.

Each egg is coded with a LOT number that identifies both the method of production and the origin of the eggs. In the first case, information on the method of laying, the storage period, the age of the hens and so on is displayed. In the second case, the information includes the country, region, and municipality of origin, as well as the specific farm. In other words, this is maximum information transparency.

To make it easier for consumers to handle the information, each product typically has a QR code that can be easily scanned with any smartphone. Once the consumer scans the code, they are redirected to a web interface where they are prompted to enter the expiry date of their product and the LOT number of the product listed on the packaging. This will display the data associated with the product in the app. All product information can then be verified by consumers directly on the blockchain.

## Conclusion

**Overall, financial cooperatives, producer and consumer cooperatives and decentralized supply chains play a key role in promoting inclusive, equitable and sustainable economic development.**

**These concepts benefit not only individuals and small businesses, but also entire communities and society. These include increased access to services, efficiency and cost reduction, elimination of middlemen, opportunities for participation in governance and greater transparency.**

# Decentralized Energy and Decentralized Finance

Of the many economic sectors where the combination of new blockchain, artificial intelligence (AI) and Internet of Things (IoT) technologies can help, one of the most interesting is the energy grid. Currently, the energy system is primarily managed by centralized entities and power generation is provided by large power plants. However, the technologies that power decentralized finance (DeFi) can also be applied to decentralized (or community) energy.

Like traditional finance, centralized energy provides some benefits to society, but it also has its drawbacks. Political conflicts and natural disasters can disrupt energy supplies or cause dramatic price increases. Centralized systems can also be highly inefficient.
In response, ways are being sought to apply DeFi platforms and digital assets to business models in the energy sector. In this article, we look at what decentralized (community) energy is and how it can find support in the world of decentralized finance and related new technologies.

# Background of Decentralized Energy

Decentralized energy systems rely on small power sources known as distributed energy resources (DERs). These systems are usually located close to the points of energy consumption and can be aggregated with each other, i.e. their capacities can be combined. This "close-to-home" approach eliminates many of the inefficiencies associated with centralized power plants, which transmit power over long distribution lines and incur significant losses.

Many DERs are renewable energy sources and can be bundled into virtual power plants, which are collections of energy resources sold as a single source. For example, the energy produced by solar panels can be bundled with hydro and wind turbine resources. A virtual power plant is therefore a group of interconnected decentralized energy sources that may be physically apart but which, from the perspective of the energy grid, act as a single entity.

The first virtual power plant projects based solely on domestic photovoltaic power plants coupled with battery energy storage are emerging around the world. For example, the Brooklyn Microgrid project is an energy marketplace using blockchain and smart contract technology that allows solar panel owners to sell their excess energy to New Yorkers. The Sonnen community uses batteries that store excess solar energy to create virtual clean energy power plants in Germany, Denmark, Spain, Italy, the United States and other countries.

The modern energy industry is moving towards decentralized generation, where a larger number of smaller units, using mainly solar, wind, hydro and geothermal energy, produce electricity instead of giant fossil fuel sources. Thus, sources of hundreds of megawatts are replacing hundreds or even units of kilowatts. The technology and operation of transmission networks are adapting to this.

Conventional power generation relies on clearly controlled electricity production and is unidirectional - from producers to consumers. The opposite is true for decentralized energy and smart grids. Its philosophy is based on the fact that distribution networks can regulate themselves (using AI technology), and every household appliance, electric car or rooftop photovoltaic system is considered part of it. The smart grid and the Internet of Things (IoT) is then intended to allow all its parts to communicate with each other using sensors and data sharing, regulate their energy

requirements, be able to store electricity and be able to supply the grid.
The hallmark of virtual power plants is the management of the resources through a common control system that optimizes and regulates the joint production of electricity in time and place according to the needs of the power grid.

## What Are the Main Benefits of Decentralized Energy?

**Improved grid efficiency**

A centralized grid loses a large amount of all the electricity it generates through transmission and distribution. These losses are partly due to the long distances that energy has to travel to get from centralized power plants to consumers.
The decentralized approach involves a collection of small power generators spread across the grid. Artificial intelligence (AI) algorithms can predict energy consumption throughout the day, and smart contracts - blockchain programs - can automate the delivery of energy where it is needed. In addition, batteries and other energy storage solutions can improve energy efficiency by ensuring that excess capacity is not wasted.

**Enhanced security**

Because of centralized energy sources, a cyber security attack, infrastructure failure or natural disaster could leave entire regions without power.
Decentralized systems are more resilient because the failure of one DER would not have a significant impact on the entire system. In addition, decentralized energy systems will reduce the country's dependence on energy imports.

**Greater transparency**

The conventional energy grid is supplied with energy from a variety of sources, including fossil fuels, nuclear power, and renewables. However, it is not possible to know how much energy comes from renewables because it is mixed with other types of energy. Virtual power plants using blockchain technology will increase transparency around energy sources.

**Lower costs**

Capital is needed to build and maintain power plants; some of this capital comes from investment banks and other financial institutions. Consumers pay part of these costs through their monthly bills. A decentralized energy system encourages businesses and individuals to participate in energy sharing without the need for centralized financial institutions, resulting in lower costs.

## Applying the Principles of Decentralized Finance and Decentralized Energy

Energy trading is an essential element of any decentralized energy system. DeFi applications can facilitate the buying and selling of energy, and tokenized digital assets can be used for energy transactions within virtual power plants. The virtual space of Web3 and DeFi is already equipped to handle the security and speed needs of the growing energy market.

DeFi protocols involved in DER management, supported by decentralized applications (dApps) used for energy resources, but also carbon credits or ESG reporting, are the future of the industry. For example, if you have a solar panel and surplus energy, you can use a smart contract to automatically distribute the surplus energy and ensure a fair distribution of the cost of the energy consumed. Ideally, decentralized energy should use blockchain technologies that are environmentally friendly, i.e., carbon neutral or carbon negative.

The combination of decentralized energy and decentralized finance brings several key benefits. Community energy enables the production of renewable energy at the local level, reduces dependence on centralized grids and promotes sustainability. DeFi, on the other hand, enables the creation of financial instruments and platforms without intermediaries, which increases their accessibility, transparency, automation, and speed.

By integrating these two concepts, individuals and communities can participate in decentralized energy projects (virtual power plants), sharing energy, costs, and rewards with each other through DeFi mechanisms, thus supporting the transition to renewable energy and the democratization of clean energy production.

# Going Forward with Regenerative Finance

**As can be seen from the above-mentioned explanations, ReFi or regenerative finance, is an area, which deals with many different tasks and topics and provides for a holistic way how to use the energy of money for common good.**

**The main principles of ReFi acknowledge the interconnectedness of social, economic, and ecological systems. The focus is on creating financial systems that are resilient and able to adapt to changing circumstances. It aims to promote social equity and inclusivity by directing financial resources towards projects and initiatives that benefit marginalized communities and disadvantaged groups.**

**Regenerative finance aims to design financial systems and investments that contribute positively to the environment and society, generating profits over time. Instead of solely focusing on short term gains and beyond measuring success solely based on financial returns, regenerative finance considers and actively seeks long-term positive impacts of financial decisions on ecosystems, communities, and future generations.**

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